

Vinay Gundamaneni

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PROFESSIONAL SUMMARY

Senior Data Engineer with 12+ years building scalable financial and operational data pipelines on Azure (Databricks, Synapse, ADF). Expert in Python, PySpark, Airflow, and Medallion architecture - with deep experience designing immutable, audit-ready ETL/ELT systems for regulated environments (HIPAA-compliant clinical and financial data). Actively integrates AI agents and LLMs into the development lifecycle: using agent-driven code generation, automated test authoring, and LLM-powered data quality checks to ship faster and with fewer defects. Proven track record migrating legacy systems to cloud-native platforms and delivering production-grade batch and streaming pipelines across finance, healthcare, and logistics.

CORE COMPETENCIES

Financial Data Pipelines

ETL/ELT Architecture

Python · PySpark · Spark

Apache Airflow

Delta Lake · Iceberg Formats

Data Immutability & Auditability

Azure Databricks · Synapse

Cloud Data Warehousing

AI Agent Integration

LLM-Powered Development

WORK EXPERIENCE

Total Quality Logistics

Sep 2022 - Present

Senior Data Engineer · Remote

- Designed **Medallion architecture** (staging → silver → gold) enforcing data immutability, quality validation, and auditability across batch and streaming financial and operational workloads
- Built financial and operational data pipelines using **Azure Data Factory and PySpark**, processing high-volume, time-sensitive datasets for multiple business units
- Spearheaded org-wide adoption of **Azure Databricks** for cloud-native data processing on **Delta Lake**; mentored team on Databricks patterns and engineering best practices
- Partnered with finance, operations, and product stakeholders to gather requirements and translate business strategy into scalable Azure data architecture
- Designed and deployed **CI/CD pipelines via Azure DevOps**; established data quality SLAs and code review standards
- Integrated **AI agents and LLM-based tooling** (Azure OpenAI, GitHub Copilot) into the development workflow - automating pipeline scaffolding, generating data quality test suites, and using agent-driven code review to reduce defect rate and accelerate delivery cycles
- Built Power BI dashboards presenting financial KPIs and operational metrics backed by Synapse and Databricks

Prime Healthcare

Apr 2021 - Sep 2022

Data Engineering Lead · Ontario, CA (Remote)

- Built audit-ready ELT pipelines with strict **data immutability, access controls, and lineage tracking** for clinical financial data - applying SOX-equivalent governance standards to HIPAA-regulated environments
- Led migration of legacy **SQL Server/SSIS** financial and clinical pipelines to **ADF + Azure Databricks**; added real-time **PySpark streaming** layer, reducing processing time and infrastructure cost
- Designed end-to-end data engineering infrastructure for large-scale, high-volume clinical and financial datasets using **Azure Synapse Analytics** and Databricks
- Developed **SQL and Python validation frameworks** as quality gates in ELT pipelines, ensuring data accuracy for finance-critical reporting and compliance; augmented with **LLM-generated test cases** for edge-case coverage
- Led cross-departmental initiative building cloud-based data pipelines across clinical, operational, and finance domains; coordinated requirements across 3 business units
- Managed Power BI dashboards with automated data refreshes and RLS configurations for financial reporting

Cognizant Technology Solutions - Client: Cargill Inc.

Apr 2019 - Apr 2021

Azure Data Engineer · Minneapolis, MN

- Designed **SSAS Tabular models and Power BI dashboards** for financial reporting across global business units; delivered structured financial data marts from Oracle/Hadoop to Azure SQL
- Engineered scalable pipelines integrating Oracle, Hadoop, and SQL Server into a unified Azure data platform serving financial and operational reporting with **SCD patterns** for historical data
- Built reusable **ETL pipeline components with Airflow-integrated CI/CD**; used Spark, Hive, and Azure tools for production-ready data transformations
- Collaborated with global teams to modernize legacy financial systems into Azure-based, scalable data environments

Primera Medical Technologies

Mar 2016 - Dec 2017

Software Engineer · Hyderabad, India

- Developed SSIS packages and SQL ETL pipelines for healthcare financial systems (Meditech, Epic Clarity EHR)
- Designed interactive Power BI dashboards for clinical and financial reporting; ensured HIPAA compliance

Porvus Tech Solutions

Aug 2012 - Feb 2016

MSBI Developer · Hyderabad, India

- Created SSIS workflows, SSRS reports, and SSAS cubes for healthcare and financial clients; designed multidimensional financial models and data warehouses

KEY PROJECTS

Medallion Data Platform - Total Quality Logistics Production

Designed and delivered end-to-end Medallion architecture (staging → silver → gold) for batch and streaming financial/operational workloads. Enforced data immutability, quality SLAs, and auditability at each layer. Became the org-wide standard for data processing.

Azure Databricks · Delta Lake · ADF · PySpark · Azure DevOps · Power BI

Legacy-to-Cloud Financial Pipeline Migration - Prime Healthcare Production

Led full migration of SQL Server/SSIS financial and clinical pipelines to ADF + Databricks. Added PySpark streaming for real-time analytics. Implemented audit trails, RLS, and Python validation frameworks meeting HIPAA-grade governance standards.

Azure Synapse · Azure Databricks · ADF · PySpark · SQL Server · HIPAA compliance

Unified Financial Data Platform - Cargill (via Cognizant) Production

Integrated Oracle, Hadoop, and SQL Server sources into a unified Azure data platform with SCD-compliant historical financial data. Delivered SSAS Tabular models and Power BI dashboards for global financial reporting.

Azure Data Factory · Spark · Hive · Azure SQL · SSAS Tabular · Airflow · Power BI

AI-Augmented Pipeline Development Workflow - Total Quality Logistics Internal Tooling

Introduced AI agents and LLM models into the data engineering development lifecycle at TQL. Used agent-driven scaffolding to auto-generate ADF pipeline templates from business requirements, LLM-powered test suite generation for data quality validation, and GitHub Copilot for accelerated PySpark development. Reduced pipeline onboarding time and improved test coverage across the Medallion architecture.

Azure OpenAI · GitHub Copilot · LLM Agents · Azure Databricks · ADF · PySpark

EDUCATION

MBA - Information Technology Services and Management

2018

Concordia University, St. Paul, MN

SKILLS

Pipeline & Orchestration: Apache Airflow, Azure Data Factory, Spark/PySpark, ETL/ELT, CI/CD (Azure DevOps, Jenkins)

Data Platforms: Azure Databricks, Azure Synapse Analytics, Azure Data Lake (ADLS), Delta Lake, Iceberg table formats, Azure SQL

Languages: Python (expert), SQL (expert)

Data Governance: Data immutability, audit trails, SCD, HIPAA compliance, data quality frameworks, Great Expectations-style validation

Cloud Warehousing: Azure Synapse / SQL DW, Azure SQL (columnar warehouse expertise transferable to Snowflake/Redshift/BigQuery)

BI & Reporting: Power BI, SSRS, SSAS Tabular, DAX, Power Automate

Infrastructure: Terraform (IaC), Azure DevOps, Jenkins, Medallion Architecture

AI & Agent Development: Azure OpenAI, GitHub Copilot, LLM-powered test generation, AI agent workflows, prompt engineering for data engineering automation